

# Technology #161: Incorporate a needle-free, hermetically sealed port with a self-sealing membrane at the catheter hub, creating a closed fluid circuit that eliminates open lumens during dialysis connections.

**Technology Domain:** fluidics\_hydraulic

**Mechanism:** PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Incorporate a needle-free, hermetically sealed port with a self-sealing membrane at the catheter hub, creating a closed fluid circuit that eliminates open lumens during dialysis connections.

**Maturity:** ENGINEERING\_DEFINITION

**Current State:** No physical prototype. Computational model and engineering definition exist.

**Key Risk:** killer experiment fails: falsification\_test\_from\_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

**Next Experiment:** first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

**Buyer Receives:** Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

**Buyer Must Build:** Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

**Transfer Posture:** Sponsored Validation

**Target Buyer:** domain-appropriate technology buyer (to be identified by portfolio owner)

*CONFIDENTIAL. See full engineering dossier for details.*